Smith's Prop. 4 sign follows $\#H(\mathbb{F}_p)$

`research/smith_serre_sign_20261016` · 2026-10-16 · protocol `PROTOCOL.md` frozen at `3be242141` before any $p = 11$ data

1. Question

Smith's construction fully succeeds on a genus-3 hyperelliptic curve H when a rational tractable subgroup S passes Prop. 2 (the trigonal map is rational) and Prop. 4 (the quotient J_H/S is the Jacobian of the plane quartic X rather than its quadratic twist: Serre's obstruction). Smith models both as fair coins. Two earlier studies found that `full_success` clusters within L-classes beyond isomorphism-class granularity (`smith_iso_granularity_20261015`) and that no 2-adic invariant of J_H explains it (`smith_class_invariant_20261015`). This study asks whether the Prop. 4 sign is biased by an L-class invariant.

2. Verdicts (frozen wording; $p = 11$ is confirmatory, $p = 7$ found the law)

prime	H1: sign law (monotone, max $\lvert r - N_1/(2p+2)\rvert \le 0.10$)	H2: clustering under (stratum, N_1, Prop. 2)
$p = 7$ (expl.)	**sign law holds**: monotone `true`, max deviation 0.06	**partial: residual clustering remains**: closes 80% of $U - C$ (stratum alone 54%, random-label control 63%); null 95% [0.439, 0.4461] vs $C = 0.4304$
$p = 11$	**sign law holds**: monotone `true`, max deviation 0.08	**partial: residual clustering remains**: closes 76% of $U - C$ (stratum alone 41%, random-label control 52%); null 95% [0.4787, 0.4803] vs $C = 0.475$

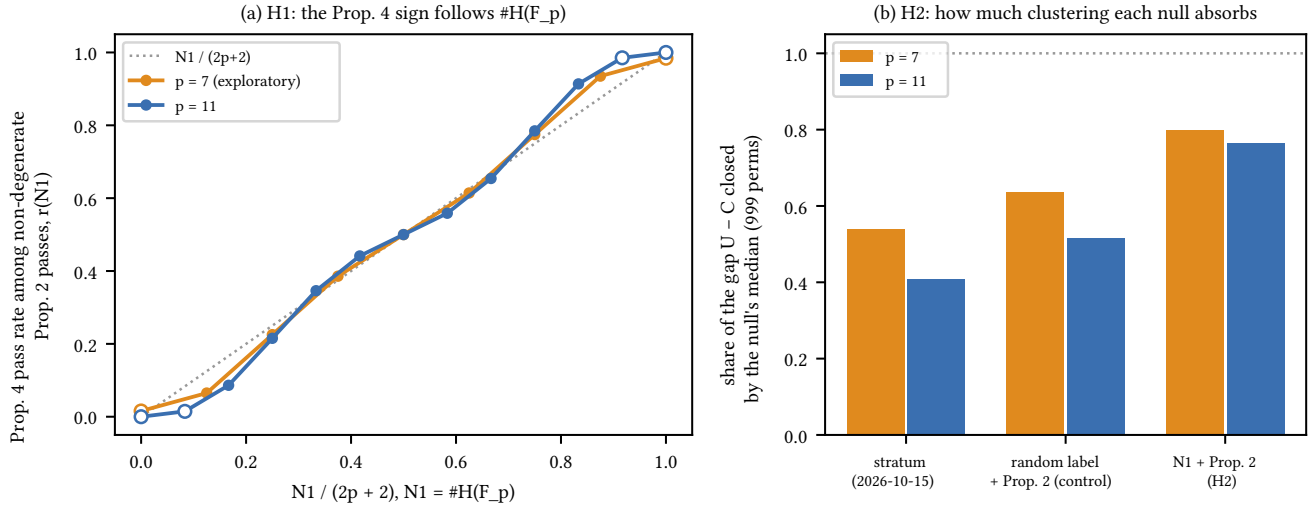


Figure 1: (a) Prop. 4 pass rate $r(N_1)$ among non-degenerate Prop. 2 passes (subgroups where exactly one of H and its twist passes Prop. 4), weighted by models, against $N_1/(2p+2)$; hollow markers are bins under 1% of the Prop. 2 passes, outside the test. (b) Share of the gap $U - C$ closed by the median of each 999-permutation null. Source `figures/plot.py`, data `data/serre_p*.json`.

3. Findings

- **Measured (both primes).** The quadratic twist preserves Prop. 2 and flips Prop. 4, except on degenerate subgroups (Smith's construction fails at small p) where Prop. 4 fails on both sides (control C5′ passes; at most 7% of Prop. 2 passes per bin at $p = 11$). So for each Prop. 2 pass the quotient curve X has either $\#X(\mathbb{F}_p) = N_1 = \#H(\mathbb{F}_p)$ (pass) or $2p + 2 - N_1$ (fail, X in the twist class).
- **Measured, frozen rule (H1).** The pass rate rises with N_1 at both primes and, rescaled by $2p + 2$, the two primes lie on one curve close to $N_1/(2p + 2)$ (Figure 1a). At $p = 11$: $N_1 = 4$: 0.086 (model 0.167); $N_1 = 6$: 0.215 (model 0.25); $N_1 = 8$: 0.346 (model 0.333); $N_1 = 10$: 0.441 (model 0.417); $N_1 = 12$: 0.5 (model 0.5); $N_1 = 14$: 0.559 (model 0.583); $N_1 = 16$: 0.654 (model 0.667); $N_1 = 18$: 0.785 (model 0.75); $N_1 = 20$: 0.914 (model 0.833). The curve is steeper than linear in the tails.
- **Consequence.** $N_1 = p + 1 + a_1$ is an L-class invariant, so Prop. 4 outcomes inside a class share a bias. A class with few rational points almost always fails Prop. 4, and its twist class almost always passes. Combined with whether any subgroup passes Prop. 2, this absorbs 80% ($p = 7$) and 76% ($p = 11$) of the gap $U - C$, against 63% / 52% for a random class label of the same coarseness. The remaining 24% at $p = 11$ is still unexplained (H2: partial).
- **Interpretation (labelled, not tested).** Between the two candidates for X, the one with more rational points is chosen roughly in proportion to its point count, as if Smith's rational data on X (the trigonal structure) were attached to a rational point. If $r \approx N_1/(2p + 2)$ persists, the bias is $\lvert a_1\rvert/(2p + 2) = O(1/\sqrt{p})$ and Smith's fair coin (Hypothesis 3) holds asymptotically. At small p it does not: Prop. 4 is a biased coin whose bias is set by the class.
- **For the walk lead.** A walk inside an L-class cannot change N_1, so it cannot escape this bias. Classes with small $\#H(\mathbb{F}_p)$ are poor walk targets, and their twist classes are good ones.

4. Validation

control	result
C1–C3 (2026-10-15 controls: census counts, invariance, partition)	all pass at both primes
C4: Prop. 2 / Prop. 4 subgroup counts constant on every isomorphism class	pass ($p = 7$: 0, $p = 11$: 0 inconsistent)
C5 (exact twist complementarity; expected to fail on degenerate subgroups)	1488 / 10251 isomorphism classes with a degenerate subgroup
C5′: curve-side + twist-side Prop. 4 passes ≤ Prop. 2 passes, per N_1 pair	`true` / `true`
random-label control (protocol)	11% extra at $p = 11$ vs 36% for N_1: under the 75% bar, so H2 is not attributed to coarse conditioning

5. Scope

Exact enumeration at $p = 7, 11$, degree-8 models. $p = 7$ was used to find the law; $p = 11$ was held out. The $N_1/(2p + 2)$ form is suggested by the data, not derived. The residual quarter of the clustering is open. Nothing here builds or prices an isogeny walk, or says anything at cryptographic sizes or in characteristic 2.

6. Reproduce

```
cargo build --release --example smith_iso_granularity
B=target/release/examples/smith_iso_granularity; D=research/smith_serre_sign_20261016/data
$B --p 7 --out $D --subgroups; $B --p 11 --out $D --subgroups   # 17 s and 16 min on 2 vCPU
$B summarize-serre --dir $D 7 11
python research/smith_serre_sign_20261016/figures/plot.py         # layout only
```

Source: B. Smith, J. Cryptology 22 (2009), arXiv:0806.2995, Props. 2 and 4, Hypothesis 3.